

Abridged Review of Multimodal Artificial Intelligence within the Understanding of Aphasia Speech

1 Introduction and Motivation

Aphasia is a common condition that affects many after experiencing a stroke. This abridged literature review supports the dissertation topic 'Aiding Communication in Aphasia Patients with Multimodal Artificial Intelligence' which explores the creation of intelligent speech aids to support the social reintegration of aphasia patients as often distanced from others and frustrated due to the inability to communicate their thoughts [1]. Traditional techniques are often inflexible and have limited personalisation towards the patient, as their needs often adapt on a daily basis, and consideration must be made for those who struggle within different aspects of communication and may benefit from different therapy techniques [2].

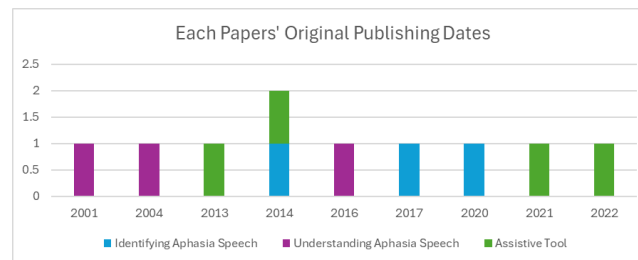


Figure 1: The original date each paper has been first published.

2 Research Field Challenges

The main challenges when working with aphasia speech are due to the lack of availability of data and commercially available Automatic Speech Recognition technology (ASRs) for disordered speech.

- **Scarcity in existing datasets** - There is large difficulty in collecting data, as people with aphasia take more time and effort to produce utterances. These small datasets restrict the use of deep neural networks and introduces the need for data augmentation which may cause the data to no longer reflect the real populations [3] [4].
- **Transcriptions of disordered speech** - Abnormal speech patterns present within aphasia outputs cannot be analysed using ASR technologies that are made for healthy speech, this means there is often necessity for manual transcription of data and labelling which requires heavy resources [3] [5][6].
- **Unpredictability within speech** - Aphasia patients with different severities and types will respond to speech therapies and rehabilitation techniques in different ways. This is since error types differ widely between language areas and will often varies within an individual overtime [7][8][5].
- **Evaluating systems** - Extra consideration must be made within the collection of system feedback from users with aphasia, to ensure both parties can be understood with ease [2].
- **Extensions to spoken language and region accents** - Although findings can be applied across languages, there are inherent restrictions into the use of produced applications and resources due to the distinct patterns of speech [8][9].

3 Scope of the Survey

Approaches to aphasia rehabilitation may focus on the recovery of the patient's communication skills, or may place focus on external changes that can make activities easier. This may be through training conversational partners to adapt to the speaker's abilities, and translating works into aphasia-friendly text [10]. The initial scope of this survey has focused on the use of multimodal models to understand aphasia speech. This includes the recognition and analysis of aphasic utterances with aims to support the speaker's communication. The scope has been expanded to include other existing AI approaches that aim to support those with aphasia due to a lack of range within the existing works.

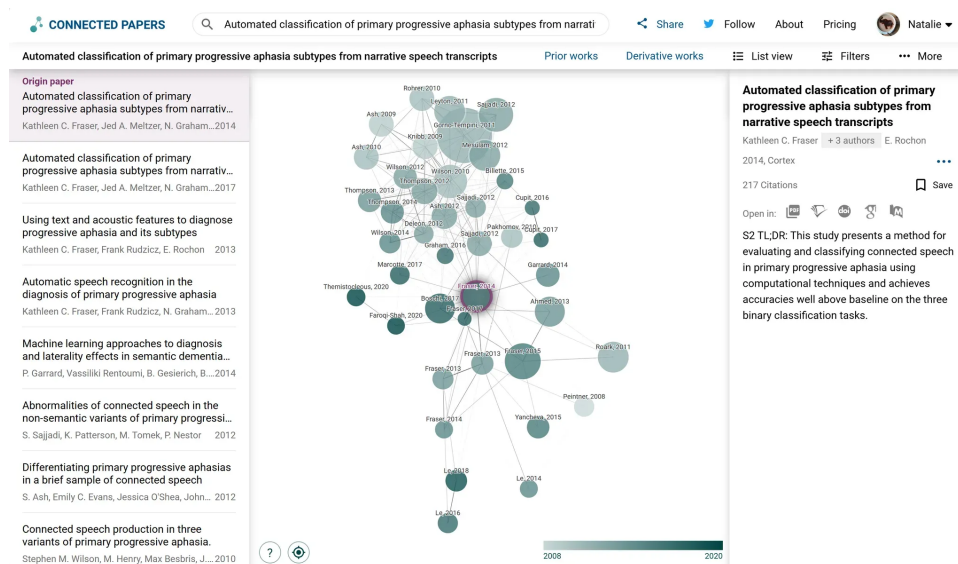


Figure 2: ConnectPapers.com visualising the connected network of Paper One. URL LINK

4 Search Methodology

In early April the dissertation supervisor, Aly Magassouba, was contacted via email and provided insight into the topic with two key questions to prompt our discussion:

Can we understand aphasic sentences? And, can we generate aphasia-friendly text automatically?

For this literature review, we focus on the first question. Magassouba has provided five papers to begin our research, including **Paper Two** and **Paper Eight**. He has also provided a survey paper that explores the use of AI within alternative communication devices to aid in aphasia rehabilitation [1], and a multimodal database that aims to ease the main research challenges within future research by providing aphasia data from video, audio, and transcripts in multiple languages [11]. These have been key reference points to begin our research, with papers being sourced through the references of the survey paper and citations from the multimodal database.

5 Classification of Literature and Organisation

The papers have been organised into three main categories based on their overall objectives, these are shown in Figure 3. Sub-categories have also been highlighted within the articles, such as the **aphasia types** that are represented in their training and testing data, as well as the **spoken language** used within the inputted dataset, and the **AI approach** that has been followed.

The papers can also be separated by the **communication task** performed by the experiment participants. These are shown within Figure 4. Each task uses different parts of the brain and therefore will have different evaluation advantages, as performance is impacted differently between aphasia types [8]. The majority of the works ask the participants to perform picture-naming tasks, with **Paper Six** and **Paper Nine** asking the participant to formulate a sentence containing this word. Instead, the other works focus on the analysis of spontaneous narrative speech.

Paper Objective	AI Approach	Type of Aphasia					
		Primary Progressive	Broca's	Wernicke's	Anomic	Global	Not specified
Identifying Aphasia Apeech	Deep Learning	Paper Three					
	Neural Network			Paper Two			
	Traditional ML	Paper One					
	ASR	Paper Three					
Understanding Aphasia Speech	Deep Learning		Paper Six				
	Neural Network				Paper Five		
	Traditional ML						
	ASR						Paper Four
AI Assistive Tools	Deep Learning		Paper Nine				
					Paper Ten		
	Neural Network						
	Traditional ML				Paper Seven		
	ASR		Paper Eight				

Figure 3: A visualisation of the paper categorisation. These are grouped by their main objectives with sub-categories to analyse the use of AI techniques. The remaining dimension shows the types of aphasia that are most commonly explored. The papers are coloured to highlight those that use multimodal models.

Language Task	Input Dataset			
	AphasiaBank	APS Corpus	WSJCAM0	Collected Data
Narrative Speech	Paper Three (multimodal)			Paper One
	Paper Nine (multimodal)			Paper Four
Confrontation Naming	Paper Six (multimodal)	Paper Eight	Paper Ten	Paper Two
				Paper Five
Sentence Formation	Paper Six (multimodal)			Paper Seven (multimodal)
	Paper Nine (multimodal)			Paper Four
				Paper Seven (multimodal)
Repetition	Paper Nine (multimodal)			Paper Two
Comprehension				Paper Two
				Paper Four

Figure 4: A visualisation of the input-data for the papers, categorised by the language tasks conducted to produce the input audio signals, and the source datasets used to train the models: AphasiaBank [11], APS Corpus [2] and WSJCAM0 [12]. The entries are coloured based on the spoken language analysed: American English, British English, Cantonese, Finish, Portuguese, and Unspecified.

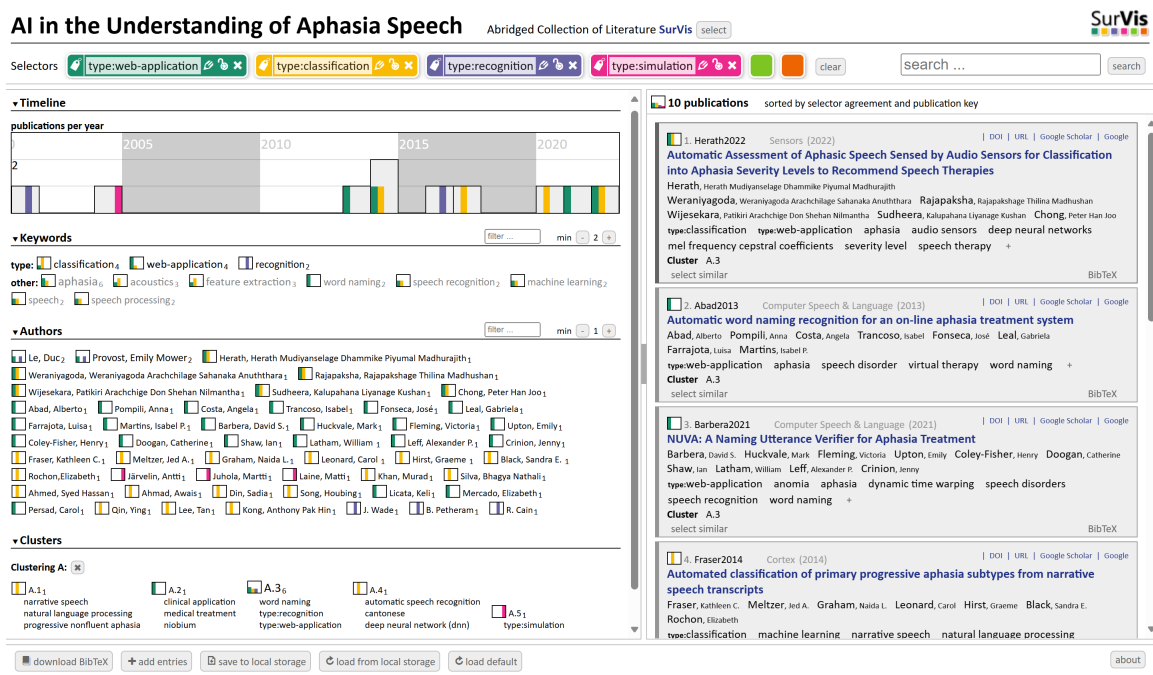


Figure 5: SurVis page of the paper collection. URL: <https://psxnr4.github.io/src/>

6 Abridged literature review

This section is organised based on the largest categorisation of the papers: their main objectives. Within the subsections the papers are ordered by AI use, discussing use of traditional machine learning technologies first, and deep learning networks at the end of the sub-sections.

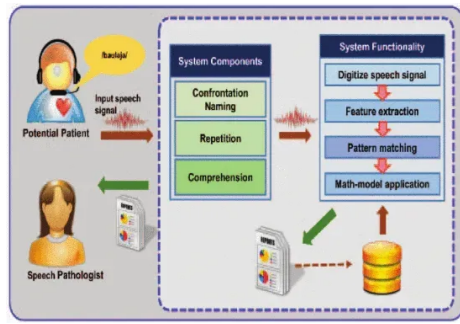
6.1 Identifying Aphasia Speech

This subsection summarises papers that use machine learning classifiers to automate the diagnosis of Aphasia.

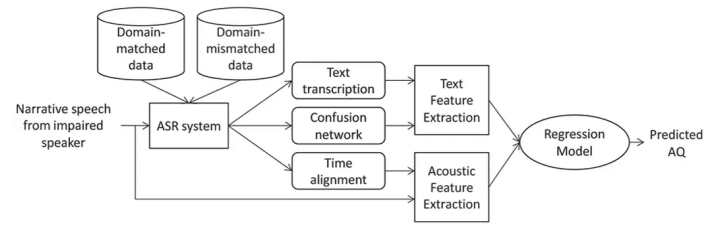
In **Paper One** [6], the authors present a diagnosis of the type of aphasia present within a patient, based on the data collected through narrative tasks and their human-generated transcripts. Syntax-analysis is performed on these and the extracted features are passed into machine learning models, Naïve Bayes and Support Vector Machines, to classify the speech. This work is built upon that of [13] and [14] who have developed machine-learning algorithms to classify transcriptions of other neurological disorders. Data is taken from three participant groups: those with semantic dementia, progressive non-fluent aphasia, and healthy controls. The performance is evaluated on the proportion of narratives that the model is able to classify with a high accuracy, using leave-one-out cross validation.

The authors of **Paper Two** [15] discuss a semi-automated system to produce a diagnostic report across two types of aphasia, based on acoustic features extracted from speech audio recordings to be adapted across different languages. The system is shown in Figure 6a; pattern matching techniques have been used to process the signal, and inferred speech characteristics help to score the patient's performance within the communication task and help predict their diagnosis. This work expands that of **Paper One** to model the interaction between text and acoustic features, and that of 'SALSA' which analyses spontaneous speech to determine other neurological disorders [16]. The model is evaluated through paired sample T-tests to analyse significance between how the different severities of the types of aphasia differ between confrontational naming and repetition tasks.

Paper Three [17] also builds upon the work of **Paper One**, to predict the severity of the condition to classify the type of Aphasia a person has based on extracted text features from ASR transcripts, using deep learning and a Siamese network. This system uses the multimodal data provided within the Cantonese AphasiaBank resource [18]. This combines text-based content from the transcripts with acoustic features, extracted from the audio recordings, to make a prediction based on both semantics and speech patterns. An overview of the system is provided within the paper, and is shown in Figure 6b. The regression model is evaluated using produced "syllable error rate" and the results are compared between that of aphasia data and the control.



(a) Paper Two



(b) Paper Three

Figure 6

6.2 Understanding Aphasia Speech

A discussion of works that explore the analysis and understanding of aphasia utterances and speaking patterns.

Paper Four [9] presents a study into the ability of existing commercial voice-interfaces to recognise and extract the content of aphasic speech compared to that of healthy control subjects. This is discussed from the perspective of using these as dictation tools to assist aphasia patients. The authors discuss the use of “speaker-adaptive” VRS [19] that requires a training period to ‘learn’ the user’s personalised speech patterns and match these to word templates to influence future speech recognition. The systems were tested on real-time audio data taken from pre-defined words and spontaneous narration of video clips. Baseline measures of software accuracy were calculated for each of the aphasia patients and non-impaired controls, after four training sessions the tests were repeated to assess potential improvements in recognition.

Paper Five [20] presents a novel dynamic model to simulate naming disorders common within aphasia speech. This is able to adapt to reflect the common changes within patients, using an MLP neural network and the addition of a randomising element to simulate naming errors. The architecture of the model is visualised within Figure 7. This processes word naming data from people with anomia from dementia or alzheimer’s disease. Existing models that simulate word naming lexilization are not able to model word learning during recovery or rehabilitation, the authors list several works as examples, such as Chialant [21] and Goldrick [22].

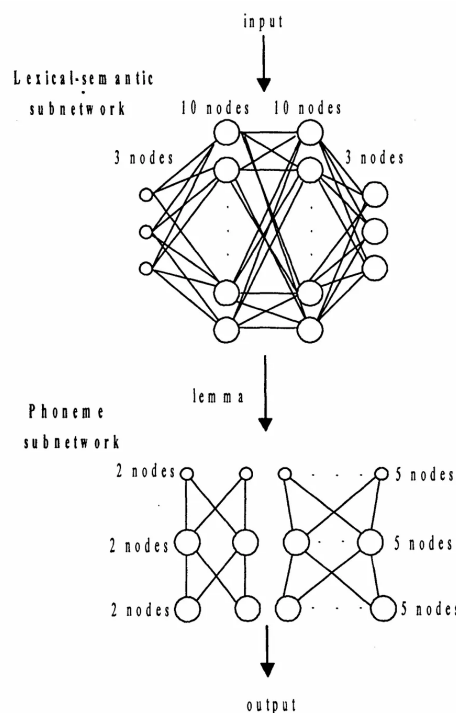


Figure 1 - Architecture of the simulation model

Figure 7: Paper Five

The authors of **Paper Six** [7] present an improved ASR system to create novel recognition of continuous disordered speech, using a Deep Neural Network and i-vector techniques. The authors build upon existing works, such as **Paper Two**, that have restricted performance of recognising aphasia speech due to the limitations of matching with acoustic models from healthy speech patterns. This is performed on the audio, visual and transcription data available in the AphasiaBank database; and the different produced adaption methods are evaluated based on the change in Production Error Rates.

6.3 Assistive Tools for Rehabilitation

A summary of the use of AI within the creation of rehabilitation tools, these use the analysis of utterances to create remote therapy exercises with personalised feedback.

The authors of **Paper Seven** [3] have developed a system, shown in Figure 8, that aims to support aphasia patients in their rehabilitation by providing personalised feedback of speech exercises outside of therapy sessions. Machine learning classifiers are used to estimate speech quality based on human validation. This is built upon the work of **Paper Eight**'s keyword spotting to recognise utterances and Black et al. who uses machine learning to estimate the reading ability of children from their speech patterns [23]. Natural speech recordings are collected from aphasia patient's describing an image using predefined word prompts. Following Black, ground-truth labels were constructed across human evaluators based on the four criteria "Clarity, Fluidity, Effort, and Prosody", but no model selection took place due to the small dataset.

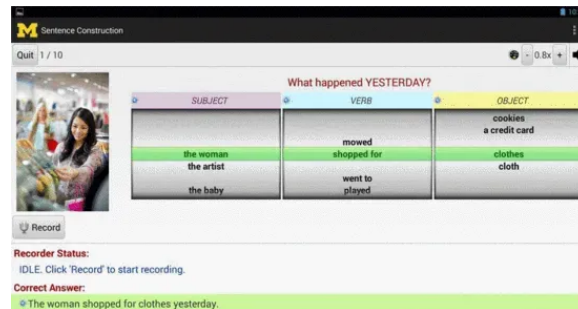
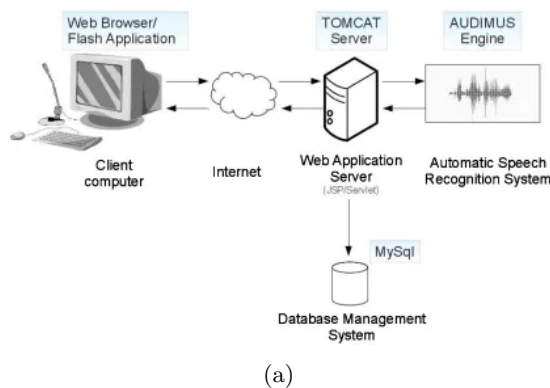


Figure 8: Paper Seven

Similarly, **Paper Eight** [2] presents the web-application 'VITHEA', shown in Figure 9a, the user is presented exercises on word recall of images and video content, and live feedback is given based on their audio responses by the "virtual therapist" to create multimodal interaction. This is a recovery tool to be used alongside conventional therapy methods, to rehabilitate aphasia patients and aim to recover their communication skills. This feedback is created through a modified ASR engine, AUDIMUS [24], using keyword spotting to validate identified spoken words similar to the work of Fraser in **Paper Two**, the process is visualised in Figure 9b. Subjective evaluation on the system takes place across hospitals and care units, receiving feedback from speech therapists to assess elements such as its appropriateness and utility.



(a)



(b)

Figure 9: Paper Eight

Paper Nine [5] presents a web-application to automate the process of identifying speech therapies that will be most beneficial to the patient. This replaces multiple clinical sessions to manually analyse the patient's speech and communication weaknesses. This work has a novel mapping of severity onto sub-categories of aphasia using supervised machine learning classification through DNNs. The system is visualised within figures 10. The author's discuss existing digital solutions for exercise practice such as 'EVA Park' [25] which creates a virtual world to socialise and practice skills, and use of sensors to both adapt a patients communication [26] and the recognition of content [27]. AphasiaBank video and audio recordings are used to train the model, that is evaluated on realtime audio using F1-score.

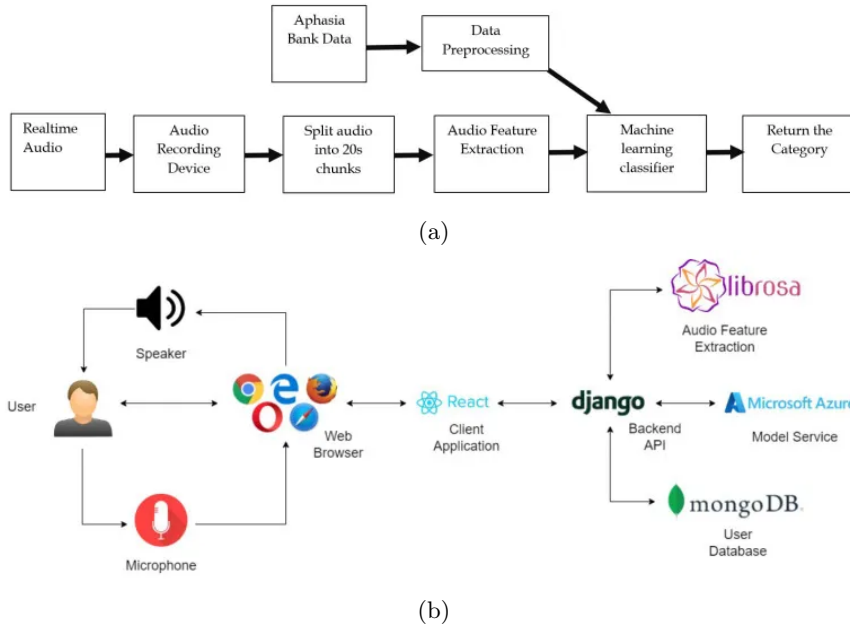


Figure 10: Paper Nine

Paper Ten [8] verifies naming exercise responses using deep learning recurrent networks. This extracts acoustic features from the responses to create a posteriorgram 11a that is compared against healthy speakers uttering the target word 11b. The authors cite two similar papers: 'Vithea' [28] an app-based validation system for Portuguese speakers and the work of Ballard using an open-source ASR-engine [29]. Though both papers report high error-rates for patients with severe impairments causing low clinical use. Live audio speech is recorded through a "gamified" app. where participants were given 6 seconds to name each item. Two clinicians independently observe the ground truth values and the system response are evaluated against this.

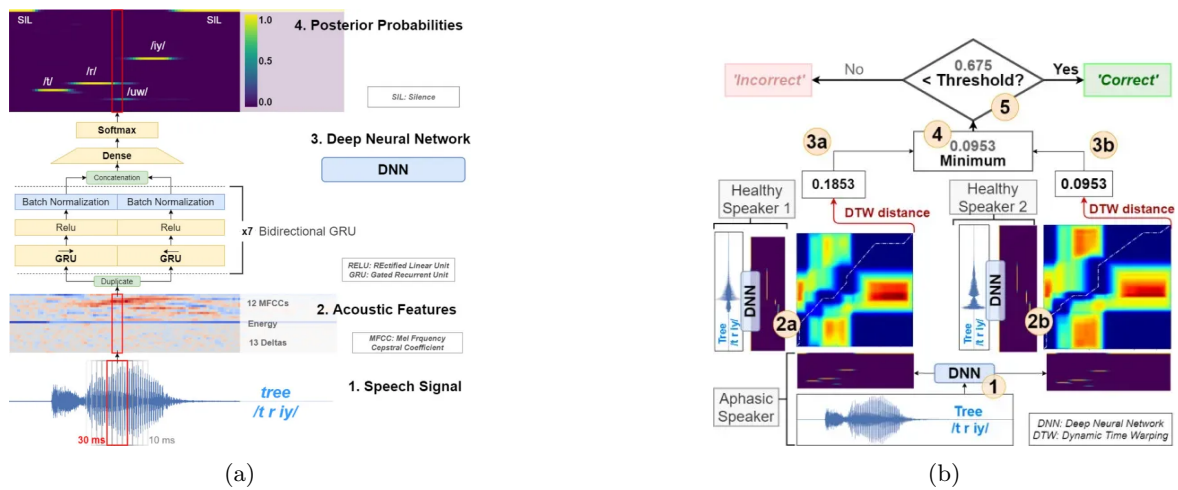


Figure 11: Paper Ten

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